

Audit Logs and Logs Explorer in GCP

Monitoring in Google Cloud Platform (GCP) refers to the set of tools and services that allow organizations to observe, collect, analyze, and act on metrics and logs generated by their applications and infrastructure. The core component for monitoring in GCP is **Cloud Monitoring** (formerly Stackdriver Monitoring), which integrates with other GCP services as well as hybrid and multi-cloud environments.

Key Features of GCP Monitoring

- **Metrics Collection:** Gathers system and application metrics from GCP services, virtual machines, Kubernetes clusters, and custom sources.
- **Dashboards:** Provides customizable dashboards for real-time data visualization.
- **Uptime Checks:** Verifies service availability from global locations.
- **Alerting Policies:** Sends alerts based on user-defined conditions, enabling proactive incident response.
- **Log Integration:** Seamlessly works with Cloud Logging to correlate metrics and logs for better context in troubleshooting.
- **Service Monitoring:** Offers out-of-the-box monitoring for GCP-native services like Compute Engine, Cloud SQL, and GKE.

Use Cases

1. **Infrastructure Monitoring**
 - Track performance and health of virtual machines, containers, and network resources.
2. **Application Performance Monitoring**
 - Monitor latency, throughput, and error rates of applications using custom and predefined metrics.
3. **Incident Management**
 - Automate alerts for critical conditions and integrate with incident response tools such as PagerDuty or Slack.
4. **Compliance and Auditing**
 - Maintain detailed logs and metrics required for compliance standards such as HIPAA, SOC 2, or ISO 27001.
5. **Capacity Planning**
 - Analyze trends to anticipate future infrastructure needs and optimize resource usage.
6. **Hybrid and Multi-cloud Monitoring**

- Extend visibility to workloads running in AWS, on-premises, or other environments using agents and APIs.

Benefits

- **Full Stack Visibility:** Offers comprehensive monitoring from infrastructure to application layer.
- **Scalability:** Natively handles large volumes of metric data across distributed systems.
- **Reduced MTTR (Mean Time to Resolution):** Accelerates detection, diagnosis, and resolution of issues.
- **Cost Optimization:** Helps identify underutilized resources and optimize spend.
- **Security and Compliance:** Provides auditable logs and monitoring to support secure operations.
- **Ease of Integration:** Works seamlessly with other GCP services and supports third-party integrations.

To begin with the Lab

1. From the GCP Console, search for IAM and navigate to it, then from the left page scroll down to Audit logs.
2. Here you can see that you have multiple logs but currently they are disabled.

The screenshot shows the GCP IAM & Admin console with the 'Audit logs' section selected in the left sidebar. The main content area displays the 'Audit logs' configuration page. At the top, there's a 'Default configuration' section with four toggle switches: 'Admin read' (Disabled), 'Data read' (Disabled), 'Data write' (Disabled), and '0 exempted principals'. Below this is the 'Data access audit logs configuration' section, which includes a filter bar and a table of services and their audit log settings.

Service	Admin read	Data read	Data write	Exempted principals	Inherited exempted principals	
☐ Access Approval				0	0	Audited meth
☐ Advisory Notifications API				0	0	Audited meth
☐ AI Platform Notebooks				0	0	
☐ AlloyDB API				0	0	Audited meth
☐ Android Device Streaming				0	0	Audited meth
☐ Anthos Multi-cloud API				0	0	Audited meth

3. Now, to enable them, click on any service to select it, for example, let's take the first service, i.e., Access Approval. Select it.
4. And here you can see that we have three permission types. We can choose any of them based on our preference.
5. You can select Exempted principals' options to exclude any of the actions of your choice.

Default configuration ⓘ

0 exempted principals

Admin read:

Disabled

Data read:

Disabled

Data write:

Disabled

Access Approval

[Permission types](#)
[Exempted principals](#)

You can configure what types of operations are recorded in your data access audit logs for the selected services. There are several subtypes of data access audit logs:

☐ **Admin read**
Records operations that read metadata or configuration information.
 ☐ **Data read**
Records operations that read user-provided data.
 ☐ **Data write**
Records operations that write user-provided data.

Save

Data access audit logs configuration

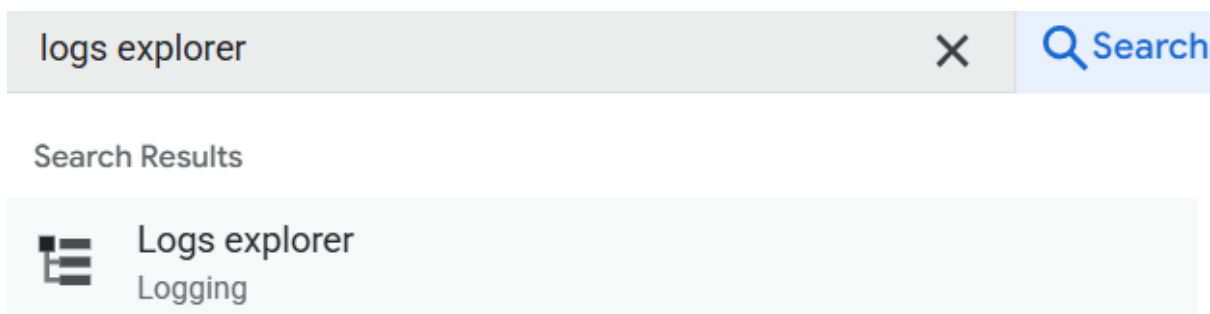
The effective data access configuration below combines the configuration for the currently selected resource and the data access configurations set on all parent resources.

Column display options...

Filter Enter property name or value ⓘ

Service ↑	Admin read	Data read	Data write	Exempted principal
☒ Access Approval				
☐ Advisory Notifications API				

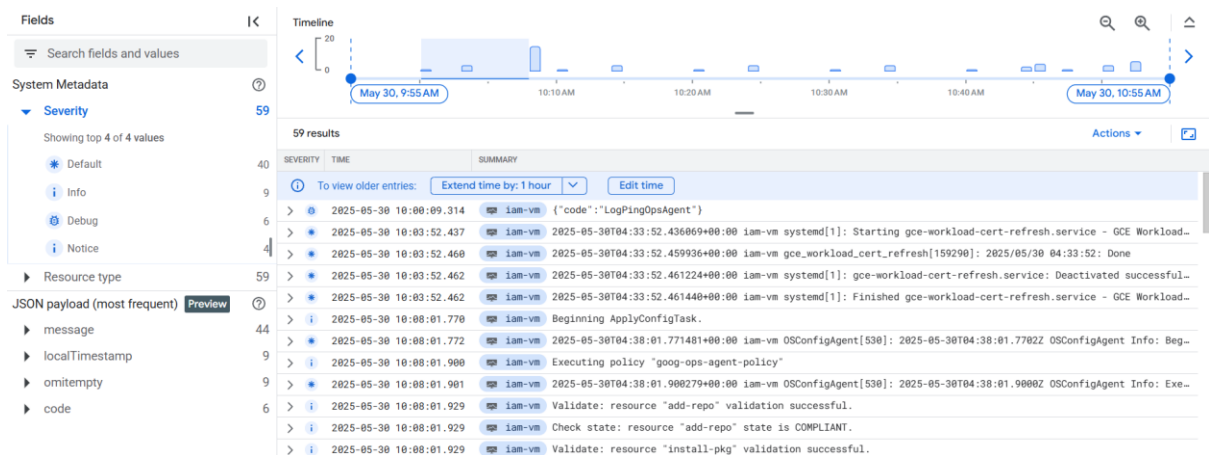
- Now, in the search bar, search for Logs Explorer and go to it.



- Now, below in the snapshot, you can see the dashboard of Logs Explorer.
- The marked area in the snapshot is where you are going to write a query and get the logs from it.

The image shows the Google Cloud Logs Explorer dashboard. At the top, there's a navigation bar with 'Logs Explorer', 'Query library', 'Share link', 'Preferences', and a time range selector set to 'Last 1 hour'. Below this is a search bar with 'Search all fields' and a 'Run query' button. A red rectangle highlights the search bar area. Below the search bar, there are filters for 'All resources', 'All log names', 'All severities', and 'Correlate by'. On the left, there's a sidebar with 'Example queries', 'Query language guide', and a list of fields including 'System Metadata', 'Severity', 'Resource type', 'JSON payload (most frequent)', and 'message'. The main area shows a timeline view with a bar chart and a table of 59 results. The table has columns for 'SEVERITY', 'TIME', and 'SUMMARY'. The first few rows show log entries from 'iam-vn' with various system messages.

- In the below snapshot, you can see the results based on the default queries running in the background. Currently, you can see 59 results, based on your cloud usage, which can be high or low.
- If you look at the left pane area, you can see the different fields



11. You can expand your logs as well, to have a better understanding of it.



12. Now I have an instance running from my previous labs. So, in the logs explorer I can view it, also, if you don't have any service running then you can create a VM.

Instances Observability Instance schedules

VM instances

Filter Enter property name or value

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
✓	iam-vm	us-central1-f			10.128.0.16 (nic0)	34.58.169.12 (nic0)	SSH

13. In the fields area in the resource type, you can see the VM instance. Click on it.

Fields

Search fields and values

System Metadata

Severity

Showing top 1 of 1 value

Default

X

Resource type

44

Showing top 1 of 1 value

VM Instance

44

14. When we select the VM from the resource type you can see that in the query editor we got the query that will help us to view logs about our VM.

Project logs

Search all fields

Run query

VM Instance

All log names

Default

Correlate by

Show query

1 severity="DEFAULT"

2 resource.type="gce_instance"

Example queries

Query language guide

Language: LQL

Fields

Search fields and values

System Metadata

Severity

Showing top 1 of 1 value

Default

X

Resource type

44

Showing top 1 of 1 value

VM Instance

40

INSTANCE ID

40

Log name

40

Timeline

May 30, 10:05 AM

10:30 AM

May 30, 11:06 AM

40 results

SEVERITY

TIME

SUMMARY

>

2025-05-30 10:50:03.387

iam-vm

2025-05-30T05:20:03.387306+00:00

iam-vm

systemd[1]: man-db.service: D-

>

2025-05-30 10:50:03.389

iam-vm

2025-05-30T05:20:03.388165+00:00

iam-vm

systemd[1]: Finished man-db.s-

>

2025-05-30 10:52:01.213

iam-vm

2025-05-30T05:22:01.212234+00:00

iam-vm

systemd[1]: Starting google-o-

>

2025-05-30 10:52:01.230

iam-vm

2025-05-30T05:22:01.229997+00:00

iam-vm

google_oslogin_nss_cache[1627-

>

2025-05-30 10:52:01.231

iam-vm

2025-05-30T05:22:01.230911+00:00

iam-vm

oslogin_cache_refresh[162716]-

>

2025-05-30 10:52:01.235

iam-vm

2025-05-30T05:22:01.235146+00:00

iam-vm

google_oslogin_nss_cache[1627-

>

2025-05-30 10:52:01.236

iam-vm

2025-05-30T05:22:01.235610+00:00

iam-vm

oslogin_nss_cache_refresh[162716]

15. In the left pane you can see that we have another property to choose from that id Instance ID. Click on it.

[Example queries](#)

[Query language guide](#)

Fields



Search fields and values

Showing top 1 of 1 value



VM Instance



▼ INSTANCE ID

44

Showing top 1 of 1 value

iam-vm

44

► Log name

44

16. You can see that in the query editor, we got another query added. In a similar way you can view logs on the logs explorer.

The screenshot shows the Google Cloud Logs Explorer interface. At the top, there is a query editor with the following query:

```
1 severity="DEFAULT"
2 resource.type="gce_instance"
3 resource.labels.instance_id="4342969404046900094"
```

Below the query editor, there are links for "Example queries" and "Query language guide". The interface is divided into two main sections: "Fields" and "Timeline".

The "Fields" section on the left shows a list of fields with checkboxes for selection. The "INSTANCE ID" field is selected, and the "Log name" field is also selected. The "Resource type" field is also selected.

The "Timeline" section on the right shows a timeline view of the logs. The timeline is centered around May 30, 10:08 AM. There are 35 results displayed in a table below the timeline.

SEVERITY	TIME	SUMMARY
>	2025-05-30 10:56:02.458	iam-vm 2025-05-30T05:26:02.45/404+00:00 iam-vm gce_workload_cert_refresh[162/2]
>	2025-05-30 10:56:02.459	iam-vm 2025-05-30T05:26:02.458783+00:00 iam-vm systemd[1]: gce-workload-cert-r-
>	2025-05-30 10:56:02.460	iam-vm 2025-05-30T05:26:02.459011+00:00 iam-vm systemd[1]: Finished gce-worklo-
>	2025-05-30 11:06:32.437	iam-vm 2025-05-30T05:36:32.436313+00:00 iam-vm systemd[1]: Starting gce-worklo-
>	2025-05-30 11:06:32.475	iam-vm 2025-05-30T05:36:32.474942+00:00 iam-vm gce_workload_cert_refresh[16338-
>	2025-05-30 11:06:32.477	iam-vm 2025-05-30T05:36:32.476357+00:00 iam-vm systemd[1]: gce-workload-cert-r-
>	2025-05-30 11:06:32.477	iam-vm 2025-05-30T05:36:32.476660+00:00 iam-vm systemd[1]: Finished gce-worklo-

17. If you want to try more than you can stop your VM and take a look at the logs.